ASSIGNMENT-8.2

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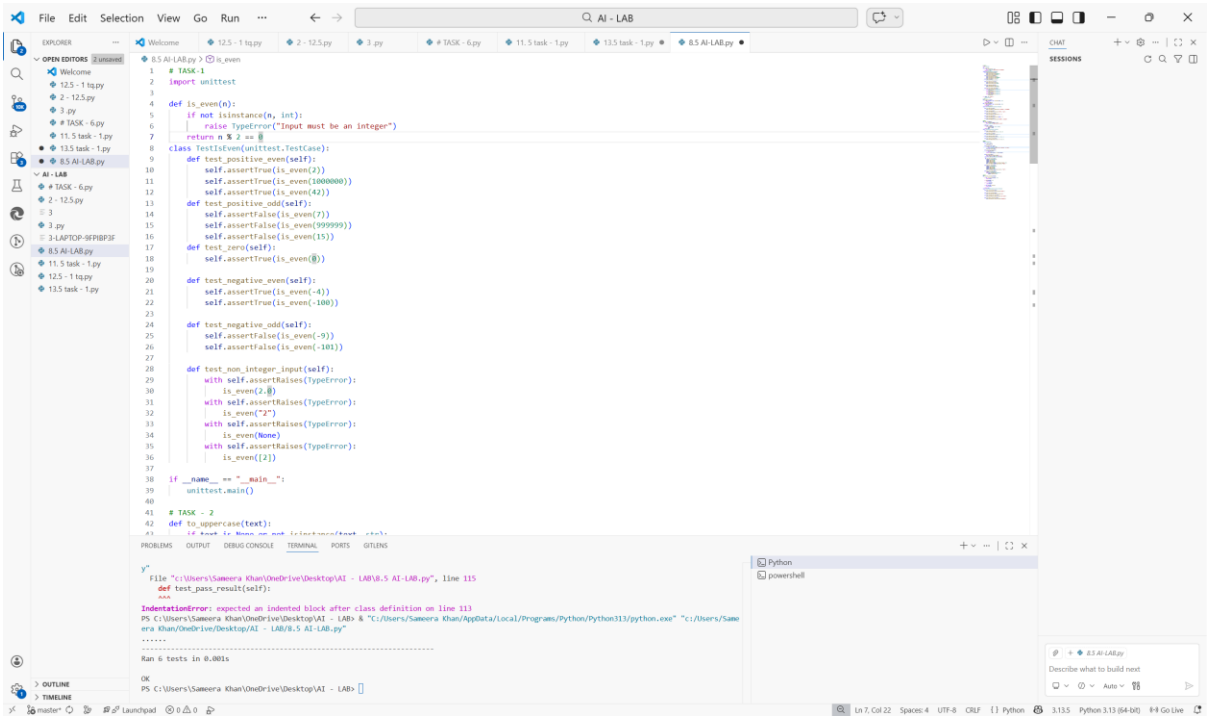
BATCH NO:29

LAB 8: TEST-DRIVEN DEVELOPMENT WITH AI – GENERATING AND WORKING WITH TEST CASES

TASK 1 – TEST-DRIVEN DEVELOPMENT FOR EVEN/ODD NUMBER VALIDATOR

CODE:

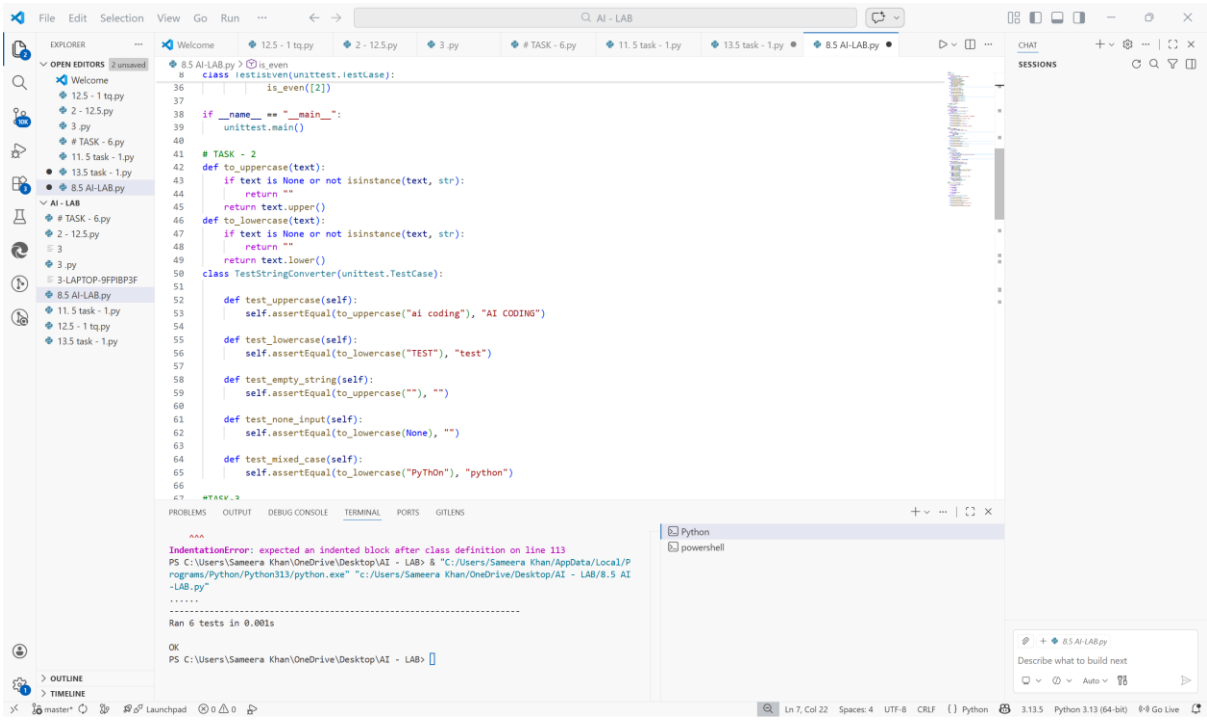


OBSERVATION:

The is_ even() function was implemented using TDD by first defining test cases for positive, negative, zero, and large integers. Input validation ensures only integers are processed, improving reliability and correctness.

TASK 2 – TEST-DRIVEN DEVELOPMENT FOR STRING CASE CONVERTER

Code:



```
class TestIsEven(unittest.TestCase):
        is_even([2])

if __name__ == "__main__":
    unittest.main()

# TASK - 2
def to_uppercase(text):
    if text is None or not isinstance(text, str):
        return ""
    return text.upper()
def to_lowercase(text):
    if text is None or not isinstance(text, str):
        return ""
    return text.lower()
class TestStringConverter(unittest.TestCase):

    def test_uppercase(self):
        self.assertEqual(to_uppercase("ai coding"), "AI CODING")

    def test_lowercase(self):
        self.assertEqual(to_lowercase("TEST"), "test")

    def test_empty_string(self):
        self.assertEqual(to_uppercase(""), "")

    def test_none_input(self):
        self.assertEqual(to_lowercase(None), "")

    def test_mixed_case(self):
        self.assertEqual(to_lowercase("PyThOn"), "python")
```

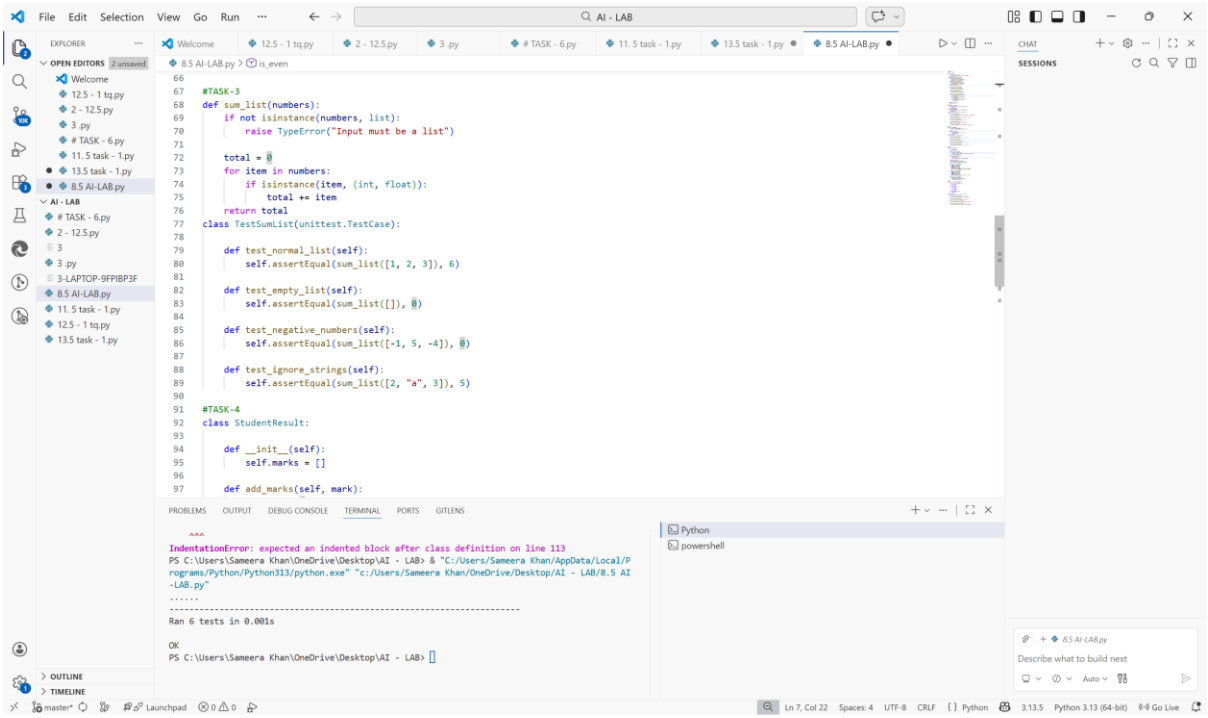
```
IndentationError: expected an indented block after class definition on line 113
PS C:\Users\Sameera Khan\OneDrive\Desktop\AI - LAB> & "C:/Users/Sameera Khan/AppData/Local/Programs/Python/Python313/python.exe" "c:/Users/Sameera Khan/OneDrive/Desktop/AI - LAB/8.5 AI-LAB.py"
.....
----------------------------------------------------------------------
Ran 6 tests in 0.001s

OK
PS C:\Users\Sameera Khan\OneDrive\Desktop\AI - LAB>
```

Observation:

Test cases were generated to verify correct conversion for mixed-case strings, empty strings, and invalid inputs. Safe input handling ensures the functions avoid runtime errors and produce consistent results.

TASK 3 – TEST-DRIVEN DEVELOPMENT FOR LIST SUM CALCULATOR

CODE:

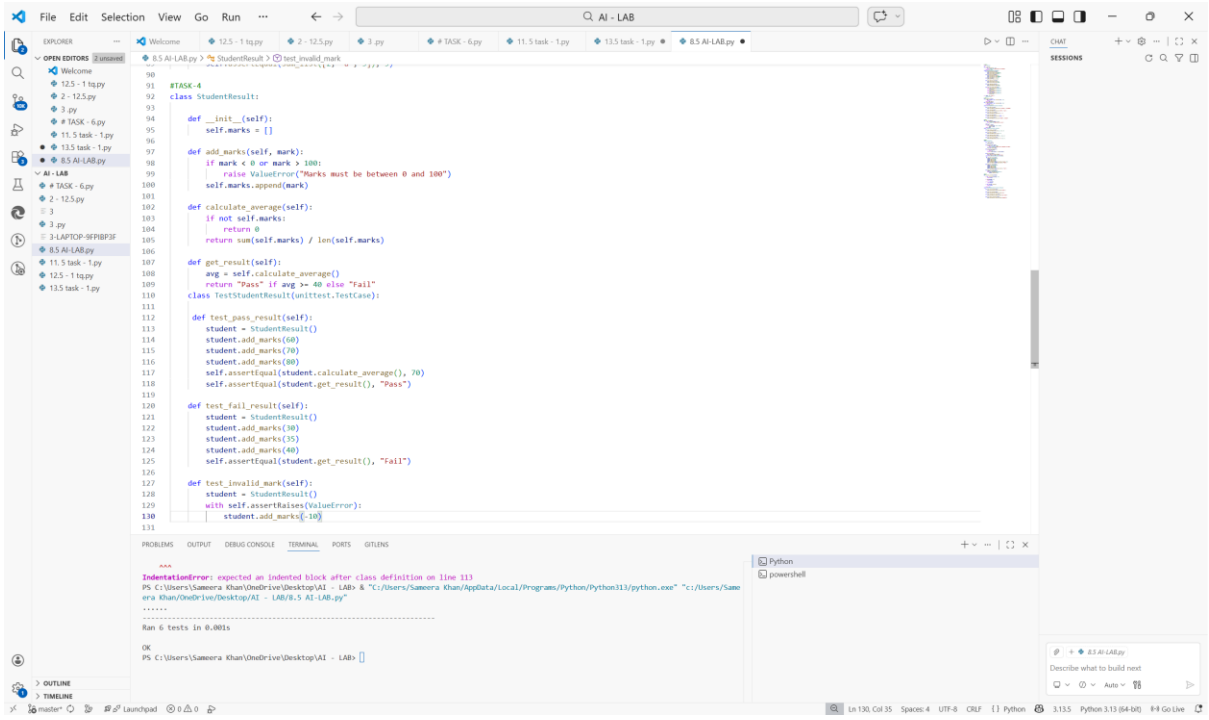


Observation:

The sum_list() function was tested with normal lists, empty lists, negative numbers, and mixed data types. Non-numeric values are ignored to ensure safe and accurate calculations.

TASK 4 – TEST CASES FOR STUDENT RESULT CLASS

CODE:

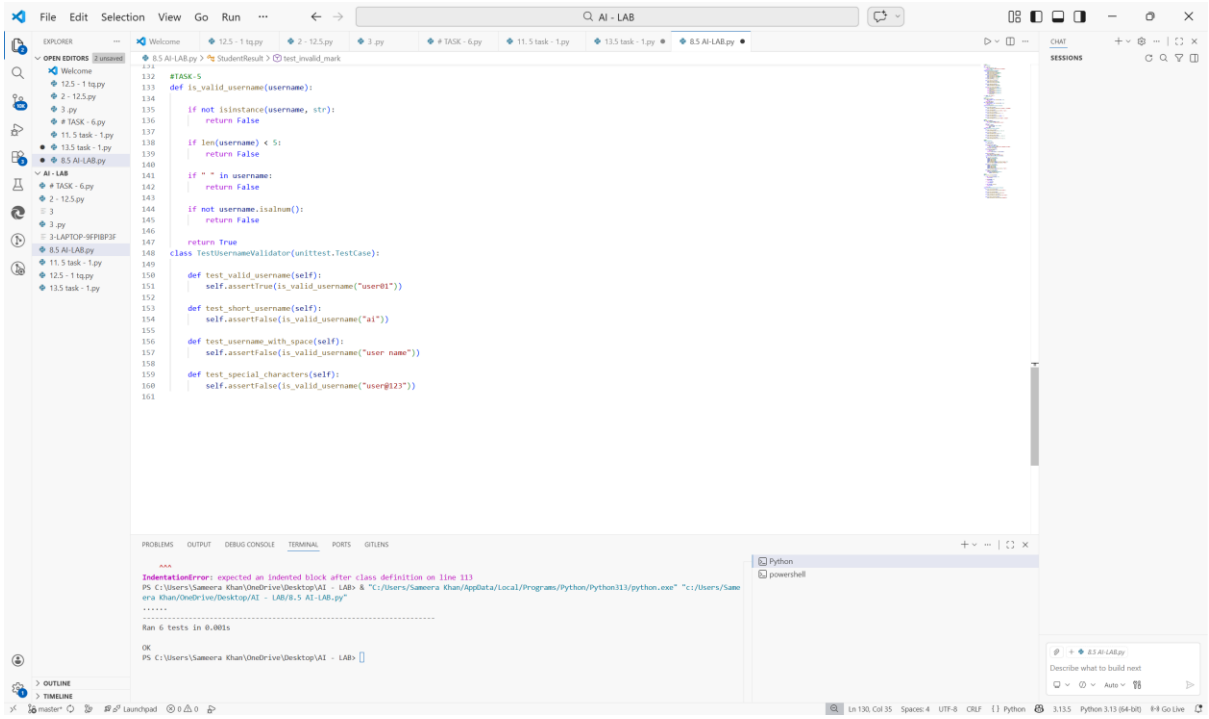


OBSERVATION:

Test cases validated the functionality of adding marks, calculating averages, and determining pass or fail results. Input validation ensures marks remain within the valid range of 0–100.

TASK 5 – TEST-DRIVEN DEVELOPMENT FOR USERNAME VALIDATOR

CODE:



OBSERVATION:

The username validation function was tested against rules such as minimum length, no spaces, and alphanumeric characters only. Test cases ensure that invalid usernames are correctly rejected while valid ones are accepted.